

AI-evolved biological factories: constraint-based evolutionary search for robust metabolic architectures in *E. coli*

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1 Introduction

Engineered microbes convert renewable feedstocks into fuels, materials, and chemicals, but the design of production strains remains guided primarily by human intuition: an engineer picks a pathway, expresses it, and hopes the host tolerates it. An alternative is to let a search process explore the space of possible metabolic architectures and select for what works, under conditions that resemble an industrial process rather than a single idealized optimum. This project asks whether a computational evolutionary search, starting from a target product rather than an existing pathway, can discover non-obvious biological architectures that match or exceed known engineered pathways under multi-condition robustness criteria. The design principles that let engineered cells maintain high productivity while remaining metabolically stable are the scientific object of interest; the evolved architecture library is the deliverable.

Three hypotheses are pre-registered (PRE-REGISTRATION.md, locked before any data or search): H1 (performance) - at least one evolved architecture reaches predicted product flux at or above the published engineered pathway evaluated in the same genome-scale model under identical uptake bounds; H2 (novelty) - the evolved library contains architectures absent from the published pathway that are mechanistically interpretable; H3 (robustness principle) - architectures selected under multi-condition pressure retain their production across a locked stress battery at a significantly higher rate than architectures selected under single-condition pressure. Honest negatives are preserved and redirect the project internally; no negative is published as the final result.

2 Pre-registered design

2.1 Hosts, models, and targets

The primary host is *Escherichia coli* K-12 MG1655 in the genome-scale metabolic model iML1515, with the yeast arm (Yeast8) reserved for cross-host robustness analysis. Three locked target products span distinct chemistry classes: 1,4-butanediol (non-natural industrial diol; benchmark route Yim et al. 2011, the Genomatica pathway), isobutanol (branched-chain higher alcohol; benchmark Atsumi et al. 2008, the keto-acid route), and lycopene (tetraterpene; benchmark Alper et al. 2005, crtEBI with dxs push). Reaction lists per the cited papers were locked in data/benchmarks.json before any search, with DOI provenance.

2.2 Benchmark metric: production envelopes

For each target, the published engineered pathway was implemented in iML1515 under the locked base condition (M9 minimal, glucose 10 mmol/gDW/h, aerobic). Because production competes with biomass, a single max-growth number is meaningless: at 100% of maximum growth all three pathways carry zero product flux. The benchmark is therefore a production envelope: maximum product flux with growth fixed at locked fractions of the FBA maximum,

$$F_t(g) = \max_v v_{\text{product}} \quad \text{s.t.} \quad Sv = 0, v_{\text{biomass}} = g \cdot \mu_{\text{max}}, l \leq v \leq u, \quad (1)$$

evaluated at $g \in \{1.0, 0.9, 0.75, 0.5, 0.25, 0.0\}$ with a parsimonious-FBA tie-break. Measured envelopes (results/benchmark_flux.json): 1,4-BDO reaches $F = 10.35$ mmol/gDW/h at $g = 0$ and 5.27 at $g = 0.5$; isobutanol 10.00 and 5.24; lycopene 0.97 and 0.49. The envelope, not a single titer, is what an evolved architecture must beat at matched growth. Table 1 reports the measured envelopes for all three published routes at the six locked growth fractions; the FBA maximum growth rate of the host under the base condition is $\mu_{\text{max}} = 0.877 \text{ h}^{-1}$ for all targets (identical uptake bounds, wild-type core). The zero at $g = 1.0$ in every row is a verified property

Table 1: Measured production envelopes (product flux, mmol/gDW/h) of the three published engineered routes in iML1515 under the locked base condition, at growth fixed to fractions of μ_{max} (results/benchmark_flux.json, run 2026-09-26; pFBA tie-break coefficient 0.999).

Target route	$g=1.0$	0.9	0.75	0.5	0.25	0.0
1,4-BDO (Yim 2011)	0.000	1.060	2.649	5.267	7.878	10.355
Isobutanol (Atsumi 2008)	0.000	1.048	2.620	5.239	7.813	10.000
Lycopene (Alper 2005)	0.000	0.098	0.244	0.488	0.731	0.970

of the host–route system, not a solver artifact: biomass and product draw on the same carbon and redox budget, so the envelopes are strictly monotone decreasing in g , and any comparison at a single growth point would be meaningless without the matched-growth rule.

2.3 Search protocol

The tie-break inside each envelope point is parsimonious FBA: among optimal solutions, minimize total flux,

$$v^* = \arg \min_v \|v\|_1 \quad \text{s.t.} \quad Sv = 0, v_{\text{product}} = F_t(g), v_{\text{biomass}} = g \mu_{\text{max}}, l \leq v \leq u, \quad (2)$$

with the pFBA optimality slack fixed at 0.999; FBA and pFBA product fluxes were verified equal on all envelope points (unique optima), so the cheaper FBA is used for search fitness and full pFBA reserved for final evaluation. The search genome encodes on/off choices over a locked reaction pool (benchmark route blocks plus literature-noted alternatives, data/reaction_pool.json). Fitness is product flux at the pFBA-optimal solution with growth fixed at $g = 0.5$, the envelope midpoint. Search fitness is the nominal-condition envelope point at the locked growth fraction,

$$\Phi(a) = F_a^{(\text{nominal})}(0.5), \quad (3)$$

and architecture performance against the benchmark route is reported as the exceedance ratio

$$E_a = \frac{F_a^{(\text{nominal})}(0.5)}{F_{\text{benchmark}}^{(\text{nominal})}(0.5)}, \quad (4)$$

with $E_a > 1$ required for an H1 beat at matched uptake bounds. The exhaustive pools used so far have genome spaces of

$$|\mathcal{G}| = 2^h \times 2^t, \quad (5)$$

genomes per target (h host-level blocks, t target-specific blocks: $h=0, t=2$ in pool v1 giving 4; $h=4, t=2$ in pool v2 giving 64), so enumeration evaluates every genome exactly once and the stochastic search budget is dominated by construction. Significance of any evolved advantage is calibrated against a 1000-seed random-architecture null as the percentile

$$\hat{q} = \frac{1 + \sum_{s=1}^{1000} \mathbf{1}[\Phi(a_s^{\text{rand}}) \geq \Phi(a)]}{1001}. \quad (6)$$

The search is a $(\mu + \lambda)$ evolutionary algorithm ($\mu = 24$, $\lambda = 48$, tournament selection, 60 generations, five locked seeds), with a 1000-seed random-architecture null for significance calibration. On the version-1 pool (two alternative blocks per target, four genomes), exhaustive enumeration strictly dominates the stochastic search and was used instead; the EA scaffold is retained for expanded pools.

2.4 Stability battery

A scalar summary of production-growth coupling used in the envelope analysis is the local slope

$$\sigma_t(g) = \frac{F_t(g_2) - F_t(g_1)}{g_2 - g_1}, \quad (7)$$

the shadow price of growth in units of product flux per unit growth fraction; the measured envelopes are strictly monotone with $\sigma_t < 0$ throughout. Robustness is evaluated over a locked nine-condition battery (nutrient shifts, carbon shifts to glycerol and acetate, anaerobiosis, uptake perturbations, oxygen limitation). After a metric-interpretation bug was caught at first evaluation (growth referenced to the base-condition maximum, which made every condition untestable), the locked rule is growth fixed at 50% of the condition-specific maximum, with retention defined against nominal-condition flux. The nine locked conditions are: GLC_LOW (glucose uptake -2), GLC_MID (-5), GLC+20% (-12), GLC -20% (-8), GLYCEROL (sole carbon, uptake -10), ACETATE (sole carbon, uptake -10), ANAEROBIC (oxygen uptake 0), O2 -50% (oxygen uptake halved), and the nominal base GLC_AEROBIC (glucose -10 , aerobic); all uptakes in mmol/gDW/h. Formally, for architecture a and condition c ,

$$R(a, c) = \frac{F_a^{(c)}(0.5)}{F_a^{(\text{nominal})}(0.5)}, \quad \text{BatteryPass}(a) = \bigwedge_{c \in \text{battery}} \mathbf{1}[R(a, c) \geq 0.8], \quad (8)$$

where $F_a^{(c)}(0.5)$ is the architecture's product flux in condition c with growth fixed at 50% of the condition-specific maximum. The pass bar of 0.8 was locked before any architecture was evaluated against the battery.

3 Results

3.1 Single-condition search reproduces the benchmark envelopes

Across all fifteen seed runs (three targets, five seeds), the best evolved architecture exactly reaches the locked envelope value at $g = 0.5$: 5.27 mmol/gDW/h for 1,4-BDO, 5.24 for isobu-

tanol, 0.49 for lycopene. The search recovers the published routes as optimal under single-condition selection - a validation of the evaluation pipeline, and the baseline for H1.

3.2 H2 candidate: a tail-only minimal 1,4-BDO architecture

A non-obvious architecture emerged from the search: with *both* heterologous upstream route blocks switched off, the four-step heterologous tail (4-hydroxybutyrate dehydrogenase, CoA transferase, CoA-acylating aldehyde dehydrogenase, alcohol dehydrogenase) still reaches the full nominal flux of 5.27 mmol/gDW/h. The model explains why: *E. coli*'s native GABA shunt (succinate-semialdehyde supply through glutamate metabolism) feeds the tail without any heterologous upstream enzyme. This tail-only architecture is absent from the Yim et al. pathway definition, uses four fewer heterologous enzymes, and is mechanistically interpretable - the H2 novelty pattern in its locked form at the nominal condition. As the tiered-battery subsection below shows, the candidate does not survive the battery, so the claim stands at nominal condition only.

3.3 H3 honest negative on pool v1, and a battery-design finding

Exhaustive multi-condition evaluation of pool v1 found that *no* genome passes the locked battery (80% retention in all nine conditions): best minimum retentions are 0.175 (1,4-BDO), 0.176 (isobutanol), and 0.27 (lycopene). Inspection shows the failure is structural, not a lack of search power: under the severe nutrient shift GLC_LOW (glucose at 2 rather than 10 mmol/gDW/h), elementary carbon balance caps a C4 product at roughly one to two mmol/gDW/h against a nominal 5.27, a retention ceiling of 0.2–0.3 that no architecture can lift. The bound is elementary: a C4 product made from glucose at uptake u_{glc} satisfies

$$v_{\text{product}} \leq \frac{6}{4} u_{\text{glc}} = 3.0 \text{ mmol/gDW/h at } u_{\text{glc}} = 2, \quad (9)$$

before any carbon is allocated to biomass, maintenance, or byproducts; with growth fixed at 50% of the condition maximum the achievable product flux falls to roughly one to two, giving $R(a, \text{GLC_LOW}) \leq 1.5/5.267 \approx 0.28$ for every conceivable architecture a - including the published routes and any architecture the search could ever return. The benchmark engineered routes fail identically. A pass bar of 80% retention on severe nutrient shifts is therefore unsatisfiable by construction, the battery provides no selection gradient, and H3 is untestable as locked. This is recorded as an honest negative with the design flaw named, and a redesign (retention against condition-achievable optimum rather than against nominal flux; tiered bars for shifts versus perturbations; robustness-oriented pool levers such as knockout blocks and co-utilization feeds) is being locked as a dated amendment before any re-evaluation, following the project's redirect-on-failure rule.

3.4 Pool v2: robustness levers shift the nominal optimum (amendment 2026-09-26)

Following the redirect rule, pool v2 was locked as a dated amendment before evaluation: four host-level robustness levers - acetate-overflow knockout (ackA-pta), formate-overflow knockout (pflB), lactate-overflow knockout (ldhA), and a glycerol co-utilization feed - added to the target-specific blocks, expanding the genome space to 64 genomes per target. Exhaustive enumeration (192 genomes total, metric-independent of the pending battery redesign) was

again strictly dominant over the stochastic budget at this size. Under the nominal condition at $g = 0.5$, the co-utilization feed dominates every target: best nominal fluxes rise to 6.496 (1,4-BDO), 6.498 (isobutanol), and 0.730 (lycopene) mmol/gDW/h, against envelope values of 5.267, 5.239, and 0.488 - nominal-condition exceedances of 23%, 24%, and 50% over the published routes evaluated in the same model. The gain is attributable to the added carbon supply, so the honest comparison under co-feeding must re-baseline the benchmark route with the same feed block enabled:

$$E_a^{\text{cofeed}} = \frac{F_a^{(\text{cofeed})}(0.5)}{F_{\text{benchmark}}^{(\text{cofeed})}(0.5)}, \quad (10)$$

with both sides evaluated under identical uptake bounds including the glycerol co-feed. That re-baselined comparison has now been executed (amendment 2026-09-27 10:04): $E^{\text{cofeed}} = 1.00$ on all three targets (6.496 vs 6.496; 6.498 vs 6.498; 0.730 vs 0.730) - the nominal exceedance is entirely attributable to the added carbon supply, and no nominal-flux advantage over the published routes survives equal-feed comparison. The 1000-draw random-architecture null likewise recovers the pool optimum on all three targets, confirming that nominal-condition search is trivial at this pool size; the project's discriminating content is the robustness structure below, not nominal flux. An honest secondary negative is recorded: for isobutanol, some knockout combinations abolish product flux entirely (minimum 0.0), so the levers are not uniformly benign even at nominal condition.

3.5 Tiered battery executed: satisfiability restored, H3 advantage unearned (amendment 2026-09-27)

The battery was redesigned and locked as a dated amendment on 2026-09-27 before evaluation: mild perturbations (glucose $\pm 20\%$, oxygen -50%) are scored as absolute retention versus nominal with a hard leg of ≥ 0.8 each, and severe nutrient shifts (GLC_LOW, GLC_MID, ANAEROBIC, GLYCEROL, ACETATE) are scored against the pool frontier in the same condition, $r_k = v_{p,k} / \max_a v_{p,k}(a)$, with a hard leg of ≥ 0.8 each; $R_{\text{tiered}} = 0.5 R_{\text{pert}} + 0.5 R_{\text{shift}}$. The bar is now satisfiable: 16/64 (1,4-BDO), 18/64 (isobutanol), and 18/64 (lycopene) genomes pass both hard legs. The best designs score $R_{\text{tiered}} = 1.0$ in all three targets: pflB knockout plus glycerol co-feed plus the succinyl-CoA route at 6.496 mmol/gDW/h nominal (1,4-BDO), pflB knockout plus co-feed plus the NADH alcohol dehydrogenase at 6.497 (isobutanol), and co-feed plus idi push at 0.730 (lycopene). Two findings follow. First, the tail-only H2 candidate does *not* survive the tiered battery ($R_{\text{tiered}} = 0.768$; severe-condition frontier-relative retention 0.41–0.67 on GLC_LOW, GLC_MID, ANAEROBIC, and ACETATE): its minimal-architecture claim stands at the nominal condition only, and the battery-robust designs are co-fed, not enzyme-minimal. Second, the H3 comparison is degenerate at this pool size: the best-by-nominal genome and the best-by- R_{tiered} genome coincide in all three targets (battery pass rates 3/3 versus 3/3), so no multi-condition search advantage is demonstrable at pool 64. This is recorded as an honest negative at current scale, not a refutation of the principle. The queued resolution was executed under the 2026-09-27 10:04 amendment: pool v3 (two further host levers, adhE and tpiA knockouts; 256 genomes per target, exhaustive) again yields $\text{argmax}(v_{\text{nominal}}) = \text{argmax}(R_{\text{tiered}})$ in all three targets (rank-1 difference 0.0000 everywhere), so the H3 search-mode advantage is not earned at N=256 either - a second falsification, reported as such. Battery pass rates at N=256: 32/256 (1,4-BDO), 60/256 (isobutanol), 22/256 (lycopene). The same amendment's H2 redirect (Robust Minimal Architecture: minimum heterologous enzyme count K_{het} subject to battery pass, published anchor $K_{\text{het}, \text{Yim}} = 5$) is also

falsified on pool v3: no battery-passing 1,4-BDO design with $K_{\text{het}} \leq 4$ exists - every robust design requires an upstream route block, so the published route's enzyme count is not beatable under the robustness constraint in this pool. The remaining live test of the robustness principle is the held-out-target arm (3-HP), locked in the same amendment.

4 Caveats

(1) All results are in-silico flux predictions; no wet-lab validation is claimed, and no build-phase engineering guidance is produced. (2) Both pools evaluated so far are deliberately small; the exhaustive results bound what these pools can express, and conclusions about search power await larger pools. In particular, the H3 multi-condition advantage is undetectable at pool 64, where the nominal-best and battery-best genomes coincide in every target. (3) The tail-only H2 candidate is verified at the nominal condition only: it fails the tiered battery, every battery-passing design requires the glycerol co-feed, and under equal-feed comparison ($E^{\text{cofeed}} = 1.00$) the co-feed fully accounts for the nominal-flux gains - no flux advantage over the published routes remains. The defensible claims are the robustness structure and its cross-target motif, not nominal flux. (4b) H3's multi-condition search advantage is falsified at both pool sizes tested (N=64 and N=256): nominal-best and battery-best designs coincide in every target. (4) GSMM predictions assume steady state and fixed biomass composition; real titers are typically lower.

5 Next steps

(1) DONE: co-feed re-baseline ($E^{\text{cofeed}} = 1.00$ on all targets) and the 1000-draw random null (recovers the optimum; nominal search trivial at this size). (2) DONE under the 10:04 amendment: pool-v3 expansion and H2-RMA redirect, both falsified as recorded above. (3) Held-out 3-HP target arm (3B, locked 10:04): construct the malonyl-CoA reductase route, run the identical tiered battery on the 64-genome host-lever pool, Fisher exact test on co-feed + pflB motif enrichment. (4) Yeast8 cross-host arm for the robustness principle.

6 Evidence appendix: matched comparisons and failed battery

6.1 Envelope as the baseline, not a titer prediction

The production-envelope values come from the committed iML1515 optimization output and are expressed as mmol/gDW/h. The six growth fractions were fixed in the design; the model’s maximum under the base condition is 0.876997 h^{-1} . These are in-silico flux optima, not measured titers, rates in a reactor, or feasible yields at every imposed growth rate. The exact source is `results/benchmark_flux.json`.

Table 2: Committed matched-growth envelope. Cells are rounded to three decimals from the JSON max-product-flux values; growth is a fraction of the model maximum.

Target	1.0	0.9	0.75	0.5	0.25	0.0
1,4-BDO	0.000	1.060	2.649	5.267	7.878	10.355
Isobutanol	0.000	1.048	2.620	5.239	7.813	10.000
Lycopene	0.000	0.098	0.244	0.488	0.731	0.970

6.2 Exhaustion of the first search space

Pool v1 has four possible genomes per target. The committed exhaustive multi-condition enumeration therefore covers the whole pool and is more decisive for this pool than stochastic algorithm rankings. The largest minimum-retention value in each pool is shown below. Every target fails the locked 0.8 retention across the nine-condition battery. That result says this battery is unsatisfiable with the modeled carbon constraint; it cannot be recast as an architecture win.

Table 3: Exhaustive pool-v1 results from `results/exhaustive_multi.json`; all battery passes are false.

Target	genomes	best minimum retention	0.8 gate
1,4-BDO	4	0.175	fail
Isobutanol	4	0.176	fail
Lycopene	4	0.270	fail

6.3 Nominal pool-v2 flux and the matched-feed caveat

The committed v2 output displays a subset of candidate genomes and identifies a total of 64 genomes per target. The maxima below are verified in the displayed rows and the manuscript’s existing v2 account; these JSON files alone do not expose all 64 individual records. The co-feed is active in the top rows, so the comparison with the published benchmark under glucose-only conditions is not feed-matched and cannot establish H1 improvement over a baseline with the same carbon supply.

Table 4: Nominal modeled flux at $g = 0.5$, results/exhaustive_nominal_v2.json; comparison benchmark comes from results/benchmark_flux.json. A matched glycerol-feed benchmark remains pending.

Target	glucose-only benchmark	pool-v2 top	top co-feed
1,4-BDO	5.267	6.496	yes
Isobutanol	5.239	6.497	yes
Lycopene	0.488	0.730	yes

Target	best retained	Retention on a common 0–1 scale
1,4-BDO	0.175	
Isobutanol	0.176	
Lycopene	0.270	
Locked threshold	0.800	

Figure 1: Each target is below the original 0.8 threshold, even at its best v1 genome. The threshold was not retroactively lowered. Source: results/exhaustive_multi.json.

6.4 Visual record of the failed v1 robustness gate

6.5 Pending gates, without imputed results

- **[PENDING]** dated battery-redesign amendment and execution against a feasible condition-matched benchmark.
- **[PENDING]** published-route baseline with the same glycerol co-feed, uptake bounds, and matched growth used in pool v2.
- **[PENDING]** independently repeated 1000-seed random-architecture null and larger reaction pool evaluation.
- **[PENDING]** Yeast8 cross-host replication and wet-lab validation; neither is claimed here.

7 Internal artifact references

1. results/benchmark_flux.json: locked flux envelopes.
2. results/exhaustive_multi.json: full v1 pool under the first battery.
3. results/exhaustive_nominal_v2.json: nominal pool-v2 subset and total counts.
4. results/battery_eval.json: condition-resolved pool-v1 checks.
5. paper/main.tex: hypotheses, methods, negatives and caveats.

8 Condition-resolved battery evidence

8.1 Interpreting retention against the nominal baseline

The locked original denominator is each architecture’s nominal glucose-aerobic flux; condition-specific growth is fixed to half the maximum in that condition. Retention above one is possible

because some perturbations improve the modeled product flux relative to the nominal condition; this does not rescue a pass requiring all conditions to exceed 0.8. The condition-resolved rows below are recorded in `results/battery_eval.json`. The three best EA architectures are compared without claiming that the EA discovered a novel robust solution.

Table 5: Condition-level retention ratios for the best pool-v1 EA architecture for each target. All entries are from the committed battery evaluation; the original 0.8 all-condition rule is unchanged.

Condition	1,4-BDO	Isobutanol	Lycopene
Low glucose	0.175	0.176	0.000
Mid glucose	0.485	0.485	0.590
Anaerobic	1.284	1.731	0.540
Glycerol	0.559	0.569	0.653
Acetate	0.243	0.245	0.000
Glucose -20%	0.794	0.794	0.836
Glucose $+20\%$	1.206	1.206	1.164
Oxygen -50%	1.000	1.000	1.000

8.2 A named comparator: tail-only 1,4-BDO

The tail-only architecture is an explicit four-step hypothesis in the main paper, not a demonstrated robust factory. The repository’s battery file records its condition-level retention separately from the nominal EA winner. Both face the same severe low-glucose and acetate constraints. The table does not infer a causal GABA-shunt mechanism beyond the model pathway accounting already discussed in the manuscript.

Table 6: Two 1,4-BDO pool-v1 designs under the same battery (`results/battery_eval.json`).

Condition	EA best	Tail-only
Low glucose	0.175	0.175
Mid glucose	0.485	0.485
Anaerobic	1.284	0.707
Glycerol	0.559	0.559
Acetate	0.243	0.243
Glucose -20%	0.794	0.794
Glucose $+20\%$	1.206	1.206
Oxygen -50%	1.000	1.000

8.3 Why this table cannot establish industrial robustness

Model retention is a steady-state flux ratio, not time-to-failure, genetic stability, titre, productivity in a fermenter, or viability after perturbation. The original condition list includes a larger carbon-input change in the low-glucose arm; the all-condition criterion is therefore structurally stringent against a fixed nominal denominator. The redesigned gate was later committed and executed separately; none of the rows above is relabeled after seeing it.

9 Locked reaction provenance and search-space size

9.1 Published-route benchmark definitions

The published-route comparator is entered as explicit reaction lists in `data/benchmarks.json`. Those citations identify pathway sources, while flux values in this manuscript come from the model run, not from physical measurements in the cited papers. The counts below are the entries in each locked pathway description; an overexpression instruction is not identical to a new heterologous reaction.

Table 7: Pathway definitions and bibliographic identifiers transcribed from `data/benchmarks.json`.

Target	listed steps	Locked benchmark citation
1,4-BDO	5	Yim et al. 2011, Nat Chem Biol 7:445-452, DOI 10.1038/nchembio.580
Isobutanol	5	Atsumi, Hanai & Liao 2008, Nature 451:86-89, DOI 10.1038/nature06450
Lycopene	4	Alper, Miyaoku & Stephanopoulos 2005, Nat Biotechnol 23:612-616, DOI 10.1038/nbt1083

9.2 What is fixed versus selectable

The pool file identifies always-on core reaction blocks separately from two selectable alternatives per target. The four-genome exhaustive v1 search is 2^2 for each target; it does not establish that the real metabolic search space has only four architectures. A block derived from an engineering search idea must not be misrepresented as a published wet-lab route.

Table 8: Searchable reaction blocks and fixed core entries from `data/reaction_pool.json`.

Target	core always on	searchable block names
14BDO	BDO2, BDO3, BDO4, BDO5	routeA_kgd, routeB_sucd
ISOBUTANOL	ALSS, KIVD	adh_nadh, adh_nadph
LYCOPENE	CRTE, CRTB, CRTI	dxs_push, idi_push

9.3 Search-design variants and attribution

- The isobutanol NADPH alcohol-dehydrogenase variant is labeled in the committed pool as a search-design engineering choice, not a literature-verified Atsumi route.
- The lycopene IDI push is likewise a search-design choice, not a literature-verified Alper route.
- The pool-v2 glycerol co-feed changes resource availability. Flux improvements under co-feeding require the same feed in the published-route comparator before attributing a win to an evolved architecture.

10 Stress battery input bounds and interpretive limits

10.1 The conditions as executed

The locked battery file `data/condition_battery.json` specifies uptake lower bounds in mmol/gDW/h; zero oxygen denotes anaerobiosis, and negative uptake bounds allow substrate import. The table lists the exact input perturbations. It is a reproducibility aid, not evidence that a laboratory cell reached those fluxes.

Table 9: Eight non-nominal stress conditions in the locked battery; nominal glucose-aerobic baseline appears in the main Methods. A dash means no explicit bound for that exchange in the condition record.

Condition	glucose	oxygen	glycerol	acetate
GLC_LOW	-2.0	-1000.0	–	–
GLC_MID	-5.0	-1000.0	–	–
ANAEROBIC	-10.0	0.0	–	–
GLYCEROL	0.0	-1000.0	-10.0	–
ACETATE	0.0	-1000.0	–	-10.0
GLC_PERT_MINUS20	-8.0	-1000.0	–	–
GLC_PERT_PLUS20	-12.0	-1000.0	–	–
O2_PERT_MINUS50	-10.0	-500.0	–	–

10.2 Why no “real-world robustness” claim follows

The low-glucose arm fixes glucose uptake to -2 rather than -10 mmol/gDW/h at nominal conditions. Its carbon limitation alone makes the same nominal-flux fraction hard to retain, independently of architecture search. The glycerol and acetate arms change the sole carbon source, while anaerobic and oxygen-halving arms alter oxygen availability. This battery measures a model’s flux ratio under those exact bounds. It does not measure serial-passage stability, process-scale reactor performance, enzyme expression, or strain viability. [PENDING: pre-locked condition-matched comparator and experimentally tractable robustness validation.]

11 Single-condition search seed ledger

11.1 Reproducibility of the matched-growth recovery

Fifteen committed search records in `results/ea_runs.jsonl` cover five seeds for each of three targets. The column below is the best modeled flux returned at the nominal midpoint $g = 0.5$, not an independent wet-lab replicate. Repetition across seeds shows the algorithm can recover a value matching the locked pathway envelope in this small pool; it does not distinguish the stochastic search from exhaustive enumeration and cannot support a new route-performance claim.

11.2 Interpretation of ties

A repeated maximum may be shared by several genomes in the same restricted reaction pool. A route-block switch that has no effect on the numerical maximum under a particular fixed-

Table 10: All fifteen single-condition runs, directly transcribed from the committed JSONL file. Flux in mmol/gDW/h; numerical agreement with the benchmark does not imply a novel design.

Target	seed	best flux	selected search blocks
14BDO	260926	5.2672	routeA_kgd, routeB_sucd
14BDO	260927	5.2672	routeA_kgd
14BDO	260928	5.2672	routeA_kgd
14BDO	260929	5.2672	routeA_kgd, routeB_sucd
14BDO	260930	5.2672	none
ISOBUTANOL	260926	5.2391	adh_nadh
ISOBUTANOL	260927	5.2391	adh_nadh, adh_nadph
ISOBUTANOL	260928	5.2391	adh_nadh
ISOBUTANOL	260929	5.2391	adh_nadh, adh_nadph
ISOBUTANOL	260930	5.2391	adh_nadh, adh_nadph
LYCOPENE	260926	0.4877	dxs_push, idi_push
LYCOPENE	260927	0.4877	dxs_push
LYCOPENE	260928	0.4877	idi_push
LYCOPENE	260929	0.4877	dxs_push, idi_push
LYCOPENE	260930	0.4877	none

growth FBA setting is not proof that the enzyme is unnecessary under every condition. Tail-only BDO remains a separate model-derived hypothesis and still fails the original all-condition robustness battery. [PENDING: a larger, independently sourced reaction pool and a matched-feed comparator evaluated on the same carbon uptake budget.]

12 Observed stress-battery retention matrix

12.1 Readout for the four tested genomes

The committed `results/battery_eval.json` records an eight-perturbation retention vector for each selected genome. This is not the original route-envelope ledger: one genome selected for lycopene has a nominal flux of 0.6127 in this battery file, whereas the published-route benchmark at $g = 0.5$ is 0.4877. Those two values have different genome contexts and must not be conflated. The battery file records product flux divided by that genome’s nominal product flux. Values above 1 are possible because a perturbation can increase the model optimum. The 80% gate is a conjunction: one failed row fails the genome.

Table 11: Every non-nominal retention entry for each genome, directly transcribed from the committed battery file. The omitted nominal baseline is 1.000 by definition.

Condition	BDO EA best	isobutanol EA best	lycopene EA best	BDO tail-only
GLC_LOW	0.175	0.176	0.000	0.175
GLC_MID	0.485	0.485	0.590	0.485
ANAEROBIC	1.284	1.731	0.540	0.707
GLYCEROL	0.559	0.569	0.653	0.559
ACETATE	0.243	0.245	0.000	0.243
GLC_PERT_MINUS20	0.794	0.794	0.836	0.794
GLC_PERT_PLUS20	1.206	1.206	1.164	1.206
O2_PERT_MINUS50	1.000	1.000	1.000	1.000

12.2 Discriminating tail-only from the benchmark under stress

The nominal BDO flux is 5.2672 for both tested architectures, but anaerobic retention is 1.284 for the EA best and 0.707 for the tail-only variant. Thus nominal equality does not imply stress equivalence. Both fail the nine-condition battery because glucose and acetate limits remain below 0.8. The high anaerobic value in the EA best is a relative model optimum, not a statement that real anaerobic production exceeds aerobic production. [PENDING: verify the flux assignments experimentally and re-test against a feasible, prospectively locked stress metric.]

12.3 Exhaustive v1 genome audit

Only two optional blocks per target were evaluated in the original pool; the four combinations per target were exhaustively tested. Fitness in the committed result is nominal flux multiplied by the minimum retention, so a nonzero fitness is not a battery pass. The table below lists all 12 genomes rather than selecting the best row. In particular, even the best minimum-retention row per target misses the required 0.8; a stochastic search over these same two bits could not find a passing genome. The lycopene bitmasks 01 and 11 have recorded zero fitness, while 10 yields

Table 12: Full v1 search space from `results/exhaustive_multi.json`; each two-bit code follows that target’s block order shown in the text. Fitness units inherit product-flux units; retention is dimensionless.

Target	block bitmask	fitness	minimum retention	all-nine pass?
14BDO: bit order routeA_kgd, routeB_sucd				
14BDO	00	0.9240	0.175	no
14BDO	01	0.9240	0.175	no
14BDO	10	0.9240	0.175	no
14BDO	11	0.9240	0.175	no
ISOBUTANOL: bit order adh_nadh, adh_nadph				
ISOBUTANOL	00	0.0000	0.000	no
ISOBUTANOL	01	0.9142	0.176	no
ISOBUTANOL	10	0.9245	0.176	no
ISOBUTANOL	11	0.9245	0.176	no
LYCOPENE: bit order dxs_push, idi_push				
LYCOPENE	00	0.0861	0.176	no
LYCOPENE	01	0.0000	0.000	no
LYCOPENE	10	0.1486	0.270	no
LYCOPENE	11	0.0000	0.000	no

the best v1 fitness of 0.1486 but only 0.270 minimum retention. The zero values should not be recast as independent failed wet-lab strains. [PENDING: a revised battery with a prospectively specified achievable reference and a larger reaction pool.]

13 Locked benchmark transformation ledger

13.1 Why each published-route baseline is auditable

`data/benchmarks.json` was locked before the first search run. The baseline is the encoded route inside iML1515 under the stated bounds, not a measured titer copied from Yim, Atsumi

or Alper. Each listed transformation below is the ledger’s description, not an independent verification of the cited paper’s experimental implementation. The entries distinguish native steps, added heterologous reactions and regulatory pushes. Where a substrate label or stoichiometric detail is missing from the source ledger, this table does not invent it.

Table 13: All 14 locked pathway-step entries. Substrate/product strings and enzyme labels are transcribed from `data/benchmarks.json`; the Alper dxs entry is a native overexpression rather than a new reaction.

Target	step	substrate	product	enzyme and recorded role
14BDO	1	alpha-ketoglutarate	succinate semialdehyde + CO ₂	alpha-ketoglutarate decarboxylase (kgd, <i>Mycobacterium</i> sp.) (listed route)
14BDO	2	succinate semialdehyde	4-hydroxybutyrate	4-hydroxybutyrate dehydrogenase (4hbD, <i>P. gingivalis</i> / <i>E. coli</i> gabD-repurposed) (listed route)
14BDO	3	4-hydroxybutyrate + acetyl-CoA	4-hydroxybutyryl-CoA + acetate	CoA transferase (cat2, <i>Porphyromonas</i> sp.) (listed route)
14BDO	4	4-hydroxybutyryl-CoA	4-hydroxybutyraldehyde + CoA	CoA-acylating aldehyde dehydrogenase (bld/ALD, <i>Clostridium</i> sp.) (listed route)
14BDO	5	4-hydroxybutyraldehyde	1,4-butanediol	alcohol dehydrogenase (adhE2, <i>Clostridium acetobutylicum</i>) (listed route)
ISOBUTANOL	1	2 pyruvate	2-acetolactate + CO ₂	acetolactate synthase (alsS, <i>Bacillus subtilis</i>) (listed route)
ISOBUTANOL	2	2-acetolactate	2,3-dihydroxy-3-methylbutanoate	ketol-acid reductoisomerase (ilvC, <i>E. coli</i> native) (native)
ISOBUTANOL	3	2,3-dihydroxy-3-methylbutanoate	2-ketoisovalerate	dihydroxy-acid dehydratase (ilvD, <i>E. coli</i> native) (native)
ISOBUTANOL	4	2-ketoisovalerate	isobutyraldehyde + CO ₂	ketoisovalerate decarboxylase (kivd, <i>Lactococcus lactis</i>) (listed route)
ISOBUTANOL	5	isobutyraldehyde	isobutanol	alcohol dehydrogenase (adhA, <i>L. lactis</i>) (listed route)
LYCOPENE	1	IPP + DMAPP + G3P-derived FPP	geranylgeranyl-PP	GGPP synthase (crtE, <i>Pantoea ananatis</i>) (listed route)
LYCOPENE	2	2 GGPP	phytoene	phytoene synthase (crtB, <i>P. ananatis</i>) (listed route)
LYCOPENE	3	phytoene	lycopene	phytoene desaturase (crtI, <i>P. ananatis</i>) (listed route)
LYCOPENE	0	pyruvate + G3P	DXP (MEP entry)	1-deoxy-D-xylulose-5-P synthase (dxs, <i>E. coli</i> native, overexpressed) (native)

13.2 Route-specific interpretation

The 1,4-BDO benchmark lists five transformations, but the search pool keeps four downstream BDO reactions always on and varies two upstream blocks. Removing both upstream blocks in the model leaves endogenous succinate-semialdehyde supply; that is a model-mechanism interpretation, not a four-enzyme strain built in the lab. The isobutanol steps 2 and 3 are annotated as native *E. coli* valine-pathway reactions, whereas its search pool’s NADPH alcohol-dehydrogenase variant is an engineering choice rather than an Atsumi route. The lycopene list has three crt enzymes and a separate native dxs push (step 0); the idi push in the search pool is not in the published-route ledger. These distinctions prevent an apparent search win from being manufactured by changing what the benchmark means. [PENDING: physical route construction and matched-growth, matched-feed measurements.]

14 Pool-v2 nominal enumeration: what was stored

14.1 The output is a top-five extract, not all 192 rows

The code in `src/exhaustive_nominal_v2.py` enumerated four binary host levers (16 combinations) times two binary target blocks (four combinations) for each of three products: 64 genome evaluations per target, 192 in all. The committed result JSON does *not* store all 192 rows. For each product it retains the five top records plus a “total 64 genomes” sentinel. Thus the paper can audit the winning settings and the reported size, but cannot reconstruct the full flux distribution, count every tie, or prove the reported worst-case flux from this truncated file alone. The five retained records per target all have glycerol co-feed enabled. Under the executed code, that switch permits glycerol uptake down to -4 mmol/gDW/h while glucose also remains available. This is additional carbon, not a controlled architecture-only performance test.

Table 14: All fifteen stored top-five rows from committed `results/exhaustive_nominal_v2.json`. The compact bit order for host levers is `ackA-pta`, `pflB`, `ldhA`, glycerol; the final two bits are the target-specific blocks. Flux is the model value at 50% maximum growth under each genome.

Target	rank in stored top five	host bits	target bits	flux
14BDO	1	0001	00	6.4956
14BDO	2	0001	01	6.4956
14BDO	3	0001	10	6.4956
14BDO	4	0001	11	6.4956
14BDO	5	0011	00	6.4956
ISOBUTANOL	1	0001	10	6.4975
ISOBUTANOL	2	0001	11	6.4975
ISOBUTANOL	3	0011	10	6.4975
ISOBUTANOL	4	0011	11	6.4975
ISOBUTANOL	5	0101	10	6.4975
LYCOPENE	1	0001	01	0.7298
LYCOPENE	2	0001	11	0.7298
LYCOPENE	3	0011	01	0.7298
LYCOPENE	4	0011	11	0.7298
LYCOPENE	5	0101	01	0.7298

14.2 The fair counterfactual was executed later

The baseline values in `results/benchmark_flux.json` used glucose uptake -10 and no additional glycerol; the v2 winners switch on glycerol. A ratio of v2 winner flux to that baseline therefore answers “what if extra feed is supplied,” not “did architecture search outperform the published route under equal feed.” The code enumerates all combinations but retains only a top-five extract per target, not the full distribution. The subsequent equal-feed published-route computation is now committed in `results/ecofeed_rebaseline_null.json` and audited below: its matched-feed ratio is 1.00 on all three targets. [PENDING: retain all 192 pool-v2 per-genome records for an independent distribution audit.]

15 Tiered battery redesign: satisfiable, but no H3 advantage

15.1 What the dated redesign replaces

The original nine-condition absolute-retention conjunction fails every pool-v1 genome under severe substrate limitation. The September 27, 09:01 locked amendment separates three mild perturbations from five severe regime shifts. The mild leg still requires each product-flux ratio against nominal to be at least 0.8. For the severe leg, each product flux is divided by the best of the 64 pool-v2 genomes *in the same target and condition*; each must also reach at least 0.8. The score is an equal-weight mean of the two leg averages, with pass requiring both hard legs. This is a different, prospective question, not retroactive passage of the old battery. The revised denominator depends on the same restricted pool, so $R_{\text{tiered}} = 1$ is a within-pool frontier position, not an industrial robustness guarantee.

15.2 Frontiers, pass counts, and the selection tie

The committed results/battery_tiered_v2.json stores five severe-condition frontiers per target, the pass count over 64 genomes, a top-five extract, and selected best-by-score and best-by-nominal genomes. It does not store all 192 per-genome vectors.

Table 15: Pool-relative severe frontiers (model flux, mmol/gDW/h) and two-hard-leg pass counts. From committed tiered battery JSON.

target	GLC low	GLC mid	anaerobic	glycerol	acetate	pass/64
14BDO	2.1641	3.7848	8.9972	2.9457	2.5205	16/64
ISOBUTANOL	2.1893	3.8048	11.8476	2.9800	2.5498	18/64
LYCOPENE	0.3270	0.4783	0.3886	0.4000	0.3603	18/64

Table 16: Best-by-tiered genomes and the exact mild/severe readouts. Every target's same genome also appears as best-by-nominal in the committed JSON, so this is a tie between selection criteria, not demonstrated H3 superiority. Genome switches are in the order ackA-pta, pflB, ldhA, glycerol, then target-specific blocks.

target	genome bits	nominal	mild minimum	severe minimum	R_{tiered}	best-mode tie?
14BDO	010101	6.4956	0.8328	1.0000	1.0000	yes
ISOBUTANOL	010110	6.4975	0.8340	0.9999	1.0000	yes
LYCOPENE	000101	0.7298	0.8624	1.0000	1.0000	yes

15.3 Interpretation and remaining evidence

At least sixteen of 64 genomes per target now pass both hard legs, so the metric has a nonzero selection gradient in this restricted pool. But the best-by-nominal and best-by- R_{tiered} choices coincide for all three targets. The locked H3 hypothesis that multi-condition selection improves pass rates over single-condition selection is therefore not demonstrated by this enumeration; no two-proportion advantage or seed-calibrated evolutionary-versus-random win is reported. The peak nominal fluxes rely on extra glycerol; the subsequently committed matched-feed published-route comparator ties them on every target. The larger-pool falsification is audited below. [PENDING: a seed-calibrated multi-condition versus single-condition search comparison and full per-genome vectors.]

16 Equal-feed and larger-pool falsification audit

16.1 Equal-feed comparator and random null

The originally reported pool-v2 nominal exceedances compared a glycerol co-fed candidate against published routes without the added feed. The later committed `results/ecofeed_rebaseline_null` makes the comparator symmetric. For 1,4-BDO, isobutanol, and lycopene respectively, the best pool-v2 nominal flux and matched-feed benchmark-route flux are 6.4956 versus 6.4956, 6.4975 versus 6.4975, and 0.7298 versus 0.7298 mmol/gDW/h. Thus $E^{\text{cofeed}} = 1.00$ for every target. This retracts the purported 23–50% architectural flux advantage: it was entirely a carbon-supply comparison artifact. A 1000-draw uniform-random genome null recovered each nominal optimum; nominal search does not beat this random comparator in the 64-genome pool. These are descriptive controls, not a new positive gate.

16.2 Pool-v3 tiered battery, second H3 negative

The locked September 27 10:04 amendment introduced two binary host levers, `ko_adhE` and `ko_tpiA`, expanding the pool to 256 genomes per target. Committed `results/battery_tiered_v3.json` records battery pass counts of 32/256, 60/256, and 22/256 for 1,4-BDO, isobutanol, and lycopene. For each target, the highest nominal-flux genome is also the highest tiered-score genome; the locked requirement for diverging selections in at least two targets is therefore not met. The same result file records no battery-passing four-or-fewer-heterologous-enzyme 1,4-BDO design: the robust-minimal-architecture H2 redirect also fails. These negatives do not erase the nominal-only tail architecture or the nonzero within-pool robustness gradients, but neither licenses a flux-win or search-superiority claim. [PENDING: held-out 3-HP motif-enrichment outcome and cross-host validation.]

17 Pool-v3 full-enumeration robustness audit

17.1 Storage permits a complete within-pool recheck

Unlike the pool-v2 top-five extract, committed `results/battery_tiered_v3.json` stores `all_scored` with 256 genome rows per target, 768 in total. Each row has the nominal flux, three mild-condition retentions, five severe-condition frontier-relative values, the two hard-leg indicators, combined pass indicator, and tiered score. These are computed model outcomes within the specified reaction pool, not biological replicates or independent strains. The per-target hard-leg counts below are recomputed from the stored Boolean rows; they agree with the top-level `n_pass` fields.

Table 17: Pool-v3 exhaustive pass ledger from all 768 stored genome rows. The two hard-leg columns count passing genomes separately; conjunction is the strict battery pass.

Target	evaluated	mild leg	severe leg	both	tiered-score range
1,4-BDO	256	128	32	32	0.7254–0.9938
Isobutanol	256	96	60	60	0–0.9997
Lycopene	256	224	22	22	0.6594–1.0000

17.2 Co-feed association is a within-pool property

All 114 battery-passers across the three targets have the glycerol co-feed enabled (32, 60, and 22 passers respectively). Yet many co-fed designs still fail: 96 for 1,4-BDO, 68 for isobutanol, and 106 for lycopene. The co-feed is therefore necessary among *this enumerated pool's* passers but not sufficient, and the pass pattern is not evidence of a feed-independent architecture improvement. The pflB knockout is present in only 16 of 32 1,4-BDO passers, 36 of 60 isobutanol passers, and 8 of 22 lycopene passers. The combined co-feed-plus-pflB motif is not universal in the pool-v3 pass set; the pre-locked held-out 3-HP enrichment test is still the appropriate prospective test of its transfer. [PENDING: held-out 3-HP pass/fail contingency table and Fisher test.]

17.3 The robust-minimal-architecture constraint has an auditable denominator

For 1,4-BDO, the stored heterologous-enzyme counts among the 32 battery passers are 16 genomes at five enzymes and 16 at six; zero have four or fewer. The published Yim-route anchor is five. Thus the proposed robust-minimal-architecture threshold of four enzymes is not reached by any passing design in this pool. The four-step tail-only design remains a nominal-condition phenomenon, not a battery-passing architecture. An all-genome enumeration resolves the within-pool question; it cannot establish that no future route or chassis could achieve the threshold. [PENDING: independent engineered-strain and cross-host validation under matched feed and measured stress conditions.]